Counterexamples for independent sets in linear regular hypergraphs

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Abstract

Cohen, Perkins, Sarantis, and Tetali conjectured an upper bound for the number of strong independent sets in a linear regular uniform hypergraph, with a corresponding bound for ordinary independent sets. We disprove the strong bound for every uniformity at least three and arbitrarily large degrees. The examples are connected and partite. For uniformity three, counterexamples exist at every even degree at least four and every odd degree at least thirteen. The ordinary-independent-set bound was previously disproved by Semchankau, Shabanov, and Shkredov; our construction gives simultaneous counterexamples to both bounds in these parameter ranges. The construction replaces each edge of a fixed hypergraph by a transversal design. We prove a counting theorem for these lifts at every fixed positive activity, in which strong and ordinary independent sets have the same second-order term. Applying the lift to high-dimensional grids and using Keevash's enumeration of multidimensional permutations also disproves the proposed strong-independent-set bound with an absolute implied constant.

1 Introduction

A set of vertices in a hypergraph is *strongly independent* if it meets each edge in at most one vertex. It is *weakly independent* if it contains no edge. The latter is the usual notion of an independent set in a hypergraph. For an r-uniform hypergraph H, write $i_2(H)$ and $i_r(H)$ for the numbers of strong and weak independent sets, respectively, including the empty set. These subscripts specify the independence condition, not the size of the sets being counted.

A hypergraph is *linear* if two distinct edges meet in at most one vertex. Throughout the paper, hypergraphs are finite and simple. A hypergraph is d-regular if every vertex lies in exactly d edges. It is r-partite if its vertex set admits a partition into r classes such that every edge meets each class in one vertex. Connectivity refers to the shadow graph, in which two vertices are adjacent when they belong to a common hyperedge. We use $\log_2$ for the binary logarithm and ln for the natural logarithm.

Cohen, Perkins, Sarantis, and Tetali [2] investigated the effect of linearity on independent-set counts. Their Conjecture 9 asserts that every n-vertex, r-uniform, d-regular linear hypergraph, with $r, d \geq 2$, satisfies

$$\frac{\log_2 i_2(H)}{n} \leq \frac{1}{r} + \frac{\log_2 r}{rd} \tag{1}$$

and

$$\frac{\log_2 i_r(H)}{n} \leq \frac{r-1}{r} + \frac{\log_2 r}{rd}. \tag{2}$$

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The weak bound (2) was already disproved by Semchankau, Shabanov, and Shkredov [6]. In particular, their Theorem 2 gives, for the three-partite addition hypergraph with three classes of size d,

$$i_3(H) = 3 \cdot 4^d + d\,3^{d+1} + O((3-c)^d)$$

for an absolute $c > 0$. The first term is precisely the proposed bound for that hypergraph, and the positive second term exceeds the error for sufficiently large d.

Our first result disproves the strong bound (1) at every uniformity $r \geq 3$. It also gives simultaneous counterexamples to the two bounds. The examples arise from rows and columns of a grid, with each grid point replaced by an independent fibre.

Theorem 1.1. *Let $r \geq 3$ be an integer and let $q \geq r+2$ be a prime. There is a connected, r-partite, linear r-uniform hypergraph $H_{r,q}$ with $n = r^2 q$ vertices and degree $d = 2q$ such that*

$$i_2(H_{r,q}) = \sum_{\ell=0}^{r} \binom{r}{\ell}^2 \ell!\,(2^q - 1)^\ell > 2^{n/r} r^{n/(rd)}. \tag{3}$$

If $q \geq 2r+3$, then also

$$i_r(H_{r,q}) > 2^{(r-1)n/r} r^{n/(rd)}.$$

For fixed r, as q tends to infinity through primes,

$$\frac{\log_2 i_j(H_{r,q})}{n} = \frac{j-1}{r} + \frac{2\log_2(r!)}{r^2 d} + o_r(d^{-1}), \qquad j \in \{2, r\}. \tag{4}$$

The second-order coefficient in (4), divided by the coefficient proposed in (1) and (2), is

$$\frac{2\log_2(r!)}{r\log_2 r} > 1.$$

This ratio tends to 2 as r tends to infinity. At uniformity three, the local designs exist at every order, and a matching deletion gives counterexamples at odd degrees as well.

Theorem 1.2. *For every even integer $d \geq 4$ and every odd integer $d \geq 13$, there is a connected, three-partite, linear, 3-uniform, d-regular hypergraph for which both (1) and (2) fail. In particular, both inequalities fail at every degree $d \geq 12$.*

The smallest instance used in our proof of Theorem 1.2 has 18 vertices, degree 4, and 352 strong independent sets. Its proposed upper bound is $192\sqrt{3}$, which is smaller than 352. We do not assert minimality of this example.

The grid construction only increases the proposed second-order coefficient by a bounded factor. A higher-dimensional grid gives a different conclusion. Conjecture 5 of [2] proposes the strong-independent-set bound

$$\frac{\log_2 i_2(H)}{n} \leq \frac{1}{r} + O\left(\frac{\log_2 r}{rd}\right). \tag{5}$$

As specified in [2], the implied constant is independent of both r and d. The following quantified statement rules out a uniform constant in (5), even when the degree is required to exceed any prescribed function of the uniformity.

Theorem 1.3. *Let $C > 0$, let $R \geq 3$ be an integer, and let $f\colon \mathbb{N} \to \mathbb{N}$ be any function. There is an integer $r \geq R$ such that, for infinitely many integers $d \geq f(r)$, a connected, r-partite, linear, r-uniform, d-regular hypergraph H satisfies*

$$\frac{\log_2 i_j(H)}{|V(H)|} > \frac{j-1}{r} + C\frac{\log_2 r}{rd}, \qquad j \in \{2, r\}.$$

Theorem 1.3 concerns a constant uniform in the uniformity. It does not rule out a bound of the form $1/r + O_r(1/d)$ for strong independent sets when r is fixed.

The construction in all three theorems uses a transversal design to replace each edge of a fixed regular linear hypergraph. Strong independent sets in the lift are counted exactly by the independent-set polynomial of the base. For weak independent sets, a container estimate and a weighted entropy inequality show that configurations with a nonempty fibre over every vertex of some base edge have exponentially smaller total weight. This yields the same second-order term for both notions of independence, at every fixed positive activity. The local counting argument builds on the container and link-counting methods used for sum-free triplets in [1, 6]; the container theorem itself is due to Saxton and Thomason [5].

The Latin-square lifting construction and local container estimates also underlie the companion manuscript [3], which determines sharp weighted residual rates and optimal counting exponents for arbitrary fixed triple-system bases. The main conclusions here are the CPST counterexamples in all uniformities and a precise counting transfer for regular bases attaining the incidence bound.

Section 2 proves the lifting and counting results. Section 3 gives the explicit grid examples and proves Theorems 1.1 and 1.2. Section 4 combines the lifting theorem with Keevash's enumeration of multidimensional permutations [4] to prove Theorem 1.3. The underlying transversal designs and multidimensional permutations are classical objects; the counterexamples depend on transferring the multiplicity of maximum independent sets from the base to its regular linear lifts.

2 Transversal-design lifts

The lifting construction separates the degree from the independent-set structure of a fixed base hypergraph. We first record the exact strong-independent-set count. We then show that, under a natural extremal-size condition on the base, the weak count has the corresponding asymptotic form.

For $j \in \{2, r\}$, define the independent-set polynomial

$$P_{A,j}(z) = \sum_{S \text{ is } j\text{-independent in } A} z^{|S|}.$$

For an activity $\lambda > 0$, write

$$Z_j(H; \lambda) = \sum_{I \text{ is } j\text{-independent in } H} \lambda^{|I|}.$$

Thus $Z_j(H; 1) = i_j(H)$.

A transversal design $\mathrm{TD}(r, q)$ is an r-partite r-uniform hypergraph with q vertices in each part, in which every pair of vertices from different parts belongs to exactly one edge. Such a design has q^2 edges and degree q. If q is a prime and $q \geq r$, one construction uses distinct elements $s_1, \ldots, s_r$ of $\mathbb{F}_q$ and the edges

$$\{(i, a + s_i b) : 1 \leq i \leq r\}, \qquad a, b \in \mathbb{F}_q.$$

Two prescribed vertices in different parts determine a, b uniquely. This also proves that the displayed edges are distinct and have the required pair property.

Let A be an r-uniform hypergraph. Replace each vertex v by a fibre V_v of size q. For every edge e of A, put a transversal design D_e of type $\mathrm{TD}(r, q)$ on the fibres indexed by e, and include no other edges. Write $L_q(A)$ for the resulting hypergraph. The designs D_e can be chosen independently and need not be isomorphic.

Lemma 2.1. *Suppose that $r \geq 3$ and that A is an N-vertex, r-uniform, k-regular linear hypergraph, where $k \geq 1$. Then $L_q(A)$ is an Nq-vertex, r-uniform, kq-regular simple linear hypergraph. Connectivity and r-partiteness of A are preserved. Moreover, for every $\lambda > 0$,*

$$Z_2(L_q(A); \lambda) = P_{A,2}((1+\lambda)^q - 1), \tag{6}$$

$$Z_r(L_q(A); \lambda) \geq P_{A,r}((1+\lambda)^q - 1). \tag{7}$$

Proof. Each vertex lies in q edges of each local design whose base edge contains its fibre, so its degree is kq. Two edges from the same local design cannot share two vertices, by uniqueness of a design edge through a pair. Two distinct base edges share at most one fibre, and a lifted edge uses only one vertex of each fibre. Edges from different local designs therefore also meet in at most one vertex. In particular, no edge is repeated.

Two vertices in distinct fibres are adjacent in the shadow of $L_q(A)$ if and only if their base vertices are adjacent in the shadow of A. There are no shadow edges within a fibre. Thus the lifted shadow is obtained by replacing each vertex of the base shadow by q independent vertices and each base shadow edge by a complete bipartite graph. A connected base has no isolated vertex because $k \geq 1$; this description proves connectivity of the lift. A partite partition of A extends by putting each fibre in the part of its base vertex.

For a lifted set I, let its support be the set of base vertices whose fibres meet I. The shadow description shows that I is strongly independent if and only if its support is strongly independent in A. A nonempty subset of one fibre has total weight $(1+\lambda)^q - 1$. Summing the product of these weights over all strong supports proves (6).

If the support is weakly independent in A, then every base edge has an empty fibre. Consequently no lifted edge is contained in I. Different supports give disjoint families, and summing their weights proves (7). □

For a regular base, strong independent sets have size at most N/r. Indeed, if S is strongly independent, counting incidences between S and the edges gives $k|S| \leq Nk/r$. Equality means that every edge meets S exactly once. A weak independent set has size at most $(r-1)N/r$, because its complement meets every edge. Equality in this second bound also means that its complement meets every edge exactly once. Hence complementation gives a bijection between strong sets of size N/r and weak sets of size $(r-1)N/r$.

The next theorem determines both weighted counts when these extremal sizes are attained. Uniformity over the choices of the local designs is part of the assertion.

Theorem 2.2. *Fix integers $r \geq 3$ and $k \geq 1$, and an N-vertex, r-uniform, k-regular linear hypergraph A. Suppose that A has a strong independent set of size N/r, and let M be the number of such sets. For every fixed $\lambda > 0$, there is a constant $c = c(A, r, \lambda) > 0$ such that*

$$Z_j(L_q(A); \lambda) = M(1+\lambda)^{(j-1)Nq/r}(1 + O_{A,r,\lambda}(\mathrm{e}^{-cq})), \qquad j \in \{2, r\}, \tag{8}$$

as q tends to infinity through orders for which $\mathrm{TD}(r, q)$ exists. The error is uniform over all choices of the designs on the base edges.

The activity, the uniformity, and the base in Theorem 2.2 are fixed before q tends to infinity. No uniformity in a varying activity or a growing base is asserted. With $n = Nq$ and $d = kq$, taking binary logarithms in (8) gives

$$\frac{\log_2 Z_j(L_q(A); \lambda)}{n} = \frac{j-1}{r}\log_2(1+\lambda) + \frac{k \log_2 M}{Nd} + o_{A,r,\lambda}(d^{-1}). \tag{9}$$

The only external input needed for the proof is the hypergraph container theorem. We state its consequence for transversal designs and check the parameters.

Lemma 2.3. *Fix $r \geq 3$ and $0 < \varepsilon < 1/2$. Every design $D = \mathrm{TD}(r, q)$ has a family $\mathcal{C}$ of subsets of $V(D)$ such that every weak independent set is contained in a member of $\mathcal{C}$, each $C \in \mathcal{C}$ satisfies $e(D[C]) \leq \varepsilon q^2$, and*

$$\ln |\mathcal{C}| = O_{r,\varepsilon}(q^{1-1/(r-1)} \ln q) = o(q).$$

The implicit constant and the threshold for q are independent of the design.

Proof. Apply Corollary 3.6 of Saxton and Thomason [5]. The design has rq vertices, average degree q, and maximum j-codegree at most 1 for every $2 \leq j \leq r$. In the notation of their codegree function, this gives

$$\delta_j \leq \frac{1}{q\tau^{j-1}}.$$

Choose $\tau = Kq^{-1/(r-1)}$. The term with $j = r$ is at most $K^{-(r-1)}$, and all the terms with $j < r$ tend to zero. Since the codegree function is a fixed positive linear combination of these terms, choosing K sufficiently large in terms of r, ε, and then q sufficiently large, gives the hypothesis $\delta(D, \tau) \leq \varepsilon/(12r!)$ and $\tau < 1/2$. The corollary supplies the covering property, the edge bound, and

$$\ln |\mathcal{C}| = O_{r,\varepsilon}(rq\tau \ln(1/\tau)).$$

Substitution of τ gives the stated estimate. All parameters used depend only on r and ε. □

The container estimate becomes useful because a sufficiently large sparse subset of a transversal design has a small part. A selected vertex in that part then imposes independent restrictions on a matching in the other parts.

Lemma 2.4. *Fix $r \geq 3$ and $\lambda > 0$. There are constants $\eta > 0$ and q_0 depending only on r, λ such that, for every $q \geq q_0$ and every $D = \mathrm{TD}(r, q)$, the total weight $B_D(\lambda)$ of weak independent sets meeting all r parts satisfies*

$$B_D(\lambda) \leq (1 + \lambda)^{(r-1)q} \mathrm{e}^{-\eta q}. \tag{10}$$

Furthermore, for some $\eta' > 0$ depending only on r, λ,

$$Z_r(D; \lambda) = r(1 + \lambda)^{(r-1)q}(1 + O_{r,\lambda}(\mathrm{e}^{-\eta' q})). \tag{11}$$

Proof. Write the parts as $V_1, \ldots, V_r$. For any $v \in V_i$, the q edges through v, with v removed, form a perfect matching on the other $(r-1)q$ vertices. Pair uniqueness proves that these sets are disjoint, and every vertex outside V_i belongs to one of them.

For a subset C of the vertices, let $c_i = |C \cap V_i|$, let $a_i = q - c_i$, and let $a = \sum_i a_i$. Each missing vertex outside V_i destroys at most one edge through a fixed vertex of $C \cap V_i$. Since every edge meets V_i once,

$$e(D[C]) \geq c_i \, [q - a + a_i]_+, \qquad [x]_+ = \max\{x, 0\}. \tag{12}$$

Set $w = 1 + \lambda$ and $b = w^{r-1} - \lambda^{r-1}$, so that $h = \ln(w^{r-1}/b) > 0$. Choose

$$\delta = \frac{1}{2(r-1)}, \qquad \gamma = \min\left\{\frac{1}{4}, \frac{h}{4 \ln w}\right\}, \qquad \varepsilon = \frac{\gamma}{4r},$$

and apply Lemma 2.3. A container of size at most $(r - 1 - \delta)q$ contributes weight at most $w^{(r-1-\delta)q}$, even if all its subsets are counted.

Suppose instead that $|C| > (r - 1 - \delta)q$. Then $a < (1 + \delta)q$. Choose i with $a_i \geq a/r$. This gives

$$q - a + a_i \geq q - \frac{r-1}{r} a > \frac{q}{2r}.$$

The sparsity of C and (12) imply $c_i < 2r\varepsilon q \le \gamma q$. A weak independent set contained in C and meeting all parts contains some $v \in C \cap V_i$. There are at most q choices of v. Its weight is λ, and arbitrary choices from the rest of the small part have weight at most $w^{\gamma q}$. In the other parts, none of the q disjoint $(r-1)$-sets in the link of v may be fully selected. Their total weight is at most b^q. Thus the contribution from this container is at most

$$q\lambda w^{\gamma q} b^q \le q\lambda w^{(r-1)q} \mathrm{e}^{-3hq/4}.$$

Both types of containers have an exponential loss relative to $w^{(r-1)q}$. Since their number is $\exp(o(q))$, summing the two bounds proves (10), with a smaller positive constant η.

Any set with an empty part is weakly independent. Inclusion–exclusion gives its exact total weight as

$$\sum_{a=1}^{r} (-1)^{a+1} \binom{r}{a} w^{(r-a)q} = r w^{(r-1)q} \bigl(1 + O_{r,\lambda}(w^{-q})\bigr).$$

Adding the contribution (10) proves (11). □

Proof of Theorem 2.2. The polynomial $P_{A,2}$ has degree N/r and leading coefficient M. Substituting $(1+\lambda)^q - 1$ in (6) proves (8) for strong independent sets. By the complementation argument preceding the theorem, $P_{A,r}$ has degree $(r-1)N/r$ and the same leading coefficient M.

It remains to bound weak independent sets whose nonempty-fibre support contains a base edge. Fix such an edge e_0. Let $\mathcal{F}$ be the family of weak independent sets in the lift that meet every fibre indexed by e_0, and let $Z_{\mathcal{F}}$ be its total weight. If this family is empty, the required bound is immediate. Otherwise choose a random member X with probability proportional to $\lambda^{|X|}$, and regard X as a vector of N fibre subsets. For a base edge e, let X_e denote its restriction to the fibres indexed by e.

Use natural logarithms in entropy. Each fibre belongs to exactly k base edges, and the entropy chain rule gives

$$k \operatorname{Ent}(X) \le \sum_{e \in E(A)} \operatorname{Ent}(X_e).$$

For completeness, order the fibre variables. In the chain-rule expansion of $\operatorname{Ent}(X_e)$, conditioning is on only the earlier variables indexed by e. Replacing them by all earlier variables can only decrease the conditional entropy. Each variable then occurs exactly k times, proving the inequality. The weight term satisfies the exact identity $\sum_e \mathbb{E}|X_e| = k\mathbb{E}|X|$. Consequently,

$$k \ln Z_{\mathcal{F}} \le \ln B_{D_{e_0}}(\lambda) + \sum_{e \ne e_0} \ln Z_r(D_e; \lambda). \tag{13}$$

Here we used $\ln Z_{\mathcal{F}} = \operatorname{Ent}(X) + (\ln\lambda)\mathbb{E}|X|$ and the elementary variational inequality $\operatorname{Ent}(Y) + (\ln\lambda)\mathbb{E}|Y| \le \ln \sum_{J \in \mathcal{S}} \lambda^{|J|}$ for a random variable supported on $\mathcal{S}$. The latter follows from nonnegativity of relative entropy with respect to the weighted distribution on $\mathcal{S}$. Since the weight identity is exact, the argument also applies when $\lambda < 1$.

There are Nk/r base edges. Applying (10) and (11) to (13) yields

$$Z_{\mathcal{F}} = O_{A,r,\lambda}\bigl((1+\lambda)^{(r-1)Nq/r} \mathrm{e}^{-\eta q/k}\bigr).$$

Taking a union bound over the fixed set of base edges gives the same estimate for all weak independent sets whose support contains a base edge. All other sets have total weight exactly $P_{A,r}((1+\lambda)^q - 1)$. Its leading term is

$$M(1+\lambda)^{(r-1)Nq/r}\bigl(1 + O_{A,r,\lambda}((1+\lambda)^{-q})\bigr).$$

Adding the exponentially smaller exceptional contribution proves the weak case of (8). The local estimates were uniform in the designs, so the final estimate has the required uniformity. □

3 Grid constructions

The row–column grid already appears in [2, Section 1.1] as the extremal example for strong independent sets at degree two. We use it as a base whose polynomial can be evaluated after lifting. We first use finite-field designs to obtain all uniformities, then use addition in a cyclic group to prescribe the degree when the uniformity is three.

Let $A(r,2)$ have vertex set $\{0,\dots,r-1\}^2$, with the r rows and the r columns as its edges. This is a connected, linear, r-uniform, 2-regular hypergraph. Assigning (i,j) to the class $i+j \pmod r$ makes it r-partite. A strong independent set is a placement of nonattacking rooks. To place ℓ rooks, choose ℓ rows, choose ℓ columns, and choose a bijection between them. Therefore

$$P_{A(r,2),2}(z) = \sum_{\ell=0}^{r} \binom{r}{\ell}^2 \ell! z^\ell. \tag{14}$$

Its maximum strong sets have size r, and there are $r!$ of them. Their complements are precisely the maximum weak sets, of size $r(r-1)$.

For a prime $q \geq r$, the finite-field lift can be written without any implicit design choices. Its vertices are

$$(i,j,z), \qquad 0 \leq i,j < r, \quad z \in \mathbb{F}_q.$$

For $a,b \in \mathbb{F}_q$, include the row and column edges

$$R_{i,a,b} = \{(i,j,a+jb) : 0 \leq j < r\}, \qquad C_{j,a,b} = \{(i,j,a+ib) : 0 \leq i < r\}. \tag{15}$$

The row indices or column indices used as field elements are distinct because $q \geq r$. Two vertices in different fibres of one row determine a,b uniquely, and the same holds in a column. Thus each row or column carries a transversal design. Lemma 2.1 proves all the asserted structural properties of this hypergraph, which we denote by $H_{r,q}$.

The following factorial estimate makes the comparison with the proposed bound uniform in r.

Lemma 3.1. *For every integer $r \geq 3$,*

$$\frac{r!}{r^{r/2}} \geq \frac{2}{\sqrt{3}}.$$

Proof. For $1 \leq i \leq r$, the paired factors satisfy $i(r+1-i) \geq r$. If r is odd, pairing all factors except $(r+1)/2$ gives

$$\frac{r!}{r^{r/2}} \geq \frac{r+1}{2\sqrt{r}} \geq \frac{2}{\sqrt{3}},$$

where the last inequality is equivalent to $(3r-1)(r-3) \geq 0$. If r is even, then $r \geq 4$. In the pairing, the pair $2, r-1$ contributes $2(r-1)$ instead of the lower bound r, while every other pair contributes at least r. Hence the ratio is at least $2(r-1)/r \geq 3/2 > 2/\sqrt{3}$. □

Proof of Theorem 1.1. Use the edges (15). Lemma 2.1 and (14) give the exact formula in (3). Since $n = r^2 q$ and $d = 2q$, the proposed strong bound is $2^{rq} r^{r/2}$. Keeping only the term with $\ell = r$ and using Lemma 3.1, we obtain

$$\frac{i_2(H_{r,q})}{2^{rq} r^{r/2}} \geq \frac{r!}{r^{r/2}} (1-2^{-q})^r \geq \frac{2}{\sqrt{3}} (1 - r2^{-q}).$$

For $q \geq r+2$, the inequality $2^{r-1} \geq r$ gives $r2^{-q} \leq 1/8$. The last expression is therefore at least $7/(4\sqrt{3}) > 1$.

There are $r!$ maximum weak supports in the base. Restricting (7) to these supports gives

$$i_r(H_{r,q}) \geq r!(2^q - 1)^{r(r-1)}.$$

The proposed weak bound is $2^{qr(r-1)} r^{r/2}$. For $q \geq 2r+3$, the elementary inequality $r(r-1) \leq 4^r$ gives $r(r-1)2^{-q} \leq 1/8$. Bernoulli's inequality and Lemma 3.1 again give a ratio at least $7/(4\sqrt{3}) > 1$.

Finally, apply (9) with base $A(r,2)$, $N = r^2$, $k = 2$, $M = r!$, and $\lambda = 1$. This proves (4) for both independence conditions. □

For triples, a cyclic group supplies the local designs at every order. Let $t \geq 2$. Replace each point of the 3×3 grid by a fibre labelled by $\mathbb{Z}/t\mathbb{Z}$. In each row and each column, take all triples whose three labels sum to zero. Any two labels determine the third, so this is a TD$(3, t)$. Denote the lift by H_t. It has $9t$ vertices, degree $2t$, and $6t^2$ edges. It is connected, linear, and three-partite by Lemma 2.1.

Put $u = 2^t$. Formula (14) gives

$$i_2(H_t) = 1 + 9(u-1) + 18(u-1)^2 + 6(u-1)^3 = 6u^3 - 9u + 4. \tag{16}$$

The proposed bound is $3\sqrt{3}\,u^3$. For $u \geq 4$, the function $6 - 9u^{-2} + 4u^{-3}$ is increasing and its value at 4 is $11/2 > 3\sqrt{3}$. Thus the strong inequality fails for every even degree $2t \geq 4$.

For the weak count, all weak supports in the base provide a useful lower bound. Inclusion–exclusion over the three full rows and the three full columns gives

$$P_{A(3,2),3}(w) = \sum_{a,b=0}^{3} (-1)^{a+b} \binom{3}{a} \binom{3}{b} w^{3(a+b)-ab} (1+w)^{9-3(a+b)+ab}.$$

Indeed, requiring a rows and b columns to be selected fixes $3(a+b) - ab$ vertices, and all other vertices may be chosen freely. Expanding yields

$$P_{A(3,2),3}(w) = 1 + 9w + 36w^2 + 78w^3 + 90w^4 + 45w^5 + 6w^6.$$

At $w = u - 1$, this becomes

$$P_{A(3,2),3}(u-1) = 6u^6 + 9u^5 - 45u^4 + 48u^3 - 18u^2 + 1 > 6u^6 \quad (u \geq 4). \tag{17}$$

To check the strict inequality, write $v = u - 4 \geq 0$. The difference from $6u^6$ is

$$3u^2(3v^3 + 21v^2 + 40v + 10) + 1 > 0.$$

By (7), the weak count is therefore larger than $6u^6$, whereas the proposed bound is $3\sqrt{3}\,u^6$.

We next arrange a perfect matching in a possibly relabelled lift. Relabelling the three fibres separately within one local design preserves its pair property and does not alter the shadow of the whole lift. It need not agree with the relabelling in another local design using the same fibre.

Lemma 3.2. *For every integer $t \geq 2$, the six local designs in a lift of $A(3,2)$ can be chosen so that the lift has a perfect matching. Deleting this matching leaves a linear, three-partite, $(2t-1)$-regular hypergraph, which is connected when $t \geq 3$.*

Proof. If t is odd, the triples $(a, a, -2a)$, for $a \in \mathbb{Z}/t\mathbb{Z}$, form a perfect matching in a row design. The three row matchings form a perfect matching of the entire lift.

Suppose that $t = 2m$ is even. In the standard addition design, the m triples $(a, a, -2a)$ with $0 \leq a < m$ form a matching, because each coordinate is distinct as a varies. Divide every

fibre into two prescribed halves of size m. For each row design, independently permute its three coordinate labels so that its displayed matching covers the first half in each fibre. For each column design, independently choose the permutations so that its displayed matching covers the second half. The six partial matchings are disjoint and cover every vertex, so their union is a perfect matching.

Deleting a perfect matching reduces every degree by one and preserves linearity and the partite partition. Before deletion, the shadow between two adjacent base fibres is $K_{t,t}$. A deleted hyperedge removes one pair from this bipartite graph, and disjoint deleted hyperedges remove disjoint pairs. Hence at most a matching is removed from each such $K_{t,t}$. For $t \geq 3$, the remaining graph is connected. In fact, two vertices on the same side have a common neighbor, since at most two of the t candidates are excluded; every vertex also has a neighbor on the opposite side. The base grid shadow is connected, so these connected bipartite graphs connect all fibres. □

Proof of Theorem 1.2. The even degrees were established by (16) and (17). For odd degrees, let $t \geq 7$ and use the lift and matching supplied by Lemma 3.2. Before deletion, the shadow and the base-support counts are those of H_t. Deleting edges cannot decrease either independent-set count. The resulting hypergraph has $n = 9t$ and $d = 2t - 1$.

After division by 2^{3t}, its strong count is at least

$$L_t = 6 - 9 \cdot 4^{-t} + 4 \cdot 8^{-t},$$

whereas the proposed bound, divided by the same factor, is

$$R_t = 3^{3t/(2t-1)}.$$

The sequence L_t is increasing and R_t is decreasing. To check the initial case, Bernoulli's inequality gives

$$\left(\frac{127}{128}\right)^{13} \geq \frac{115}{128} = \frac{7360}{8192} > \frac{6561}{8192} = \left(\frac{R_7}{6}\right)^{13}.$$

Thus $R_7 < 381/64$, whereas $L_7 > 6 - 9/16384 > 381/64$. Monotonicity proves $L_t > R_t$ for every $t \geq 7$. The weak count is greater than $6 \cdot 2^{6t}$ by (17), while its proposed bound is $2^{6t} R_t < 6 \cdot 2^{6t}$. Thus both bounds fail for every odd $d = 2t - 1 \geq 13$. All structural claims follow from Lemma 3.2. □

Remark 3.3. These lifts need not be cross-edge-free. In the terminology of [2], a cross-edge at v is an edge not containing v that contains two neighbors of v. For example, in a local addition design of order $t \geq 2$, fix v in the first part and choose an edge whose first-part vertex is different from v. Its vertices in the second and third parts are both neighbors of v. Thus it is a cross-edge at v. The counterexamples in this section do not contradict the results of [2] that assume cross-edge freeness.

4 High-dimensional bases

The same lift turns a large family of maximum strong independent sets into a large second-order term. We use the coordinate-line hypergraphs associated with Hamming graphs, a connection already discussed in [2, Section 1.1]. Keevash's enumeration of multidimensional permutations supplies the multiplicity needed to make the normalized correction arbitrarily large.

For integers $r \geq 3$ and $k \geq 2$, let $A(r, k)$ have vertex set $\{0, \ldots, r-1\}^k$. An edge is a coordinate line, obtained by fixing all but one coordinate. The hypergraph has $N = r^k$

vertices and degree k. Two distinct coordinate lines meet in at most one point, so it is linear. Changing coordinates one at a time connects any two vertices in its shadow. The classes defined by the sum of the coordinates modulo r form an r-partite partition.

The lines in any one coordinate direction partition the vertex set into r^{k-1} edges. Consequently every strong independent set has size at most r^{k-1}. The partite classes attain this bound. A set of that size is strongly independent if and only if it meets every coordinate line exactly once. Such sets are the supports of $(k-1)$-dimensional permutations of order r. Let their number be $M(r,k)$.

Keevash [4, Theorem 1.8] proved that, for each fixed positive integer s, the number of s-dimensional permutations of order r is

$$\left(\frac{r}{\mathrm{e}^s}+o(r)\right)^{r^s} \qquad \text{as } r\longrightarrow\infty.$$

Taking $s=k-1$ gives, for each fixed k,

$$\log_2 M(r,k)=r^{k-1}\bigl(\log_2 r-(k-1)\log_2 \mathrm{e}+o_k(1)\bigr). \tag{18}$$

The order of the parameters in this statement matters. The dimension k is fixed before r tends to infinity.

Proof of Theorem 1.3. Fix C,R,f as in the statement. Choose an integer $k\ge 2$ with $k>C$. By (18),

$$\frac{k\log_2 M(r,k)}{r^{k-1}\log_2 r}\longrightarrow k \qquad \text{as } r\longrightarrow\infty.$$

Choose and fix an integer $r\ge R$ large enough that this ratio is greater than C. From this point onward, both r and the base $A(r,k)$ are fixed.

For every prime $q\ge r$, take a transversal-design lift $L_q(A(r,k))$. It has $n=r^kq$ vertices and degree $d=kq$, and it is connected, r-partite, and linear by Lemma 2.1. Apply (9) with $N=r^k$, $M=M(r,k)$, and $\lambda=1$. For both $j=2$ and $j=r$,

$$\lim_{\substack{q\to\infty\\ q\text{ prime}}}\frac{rd}{\log_2 r}\left(\frac{\log_2 i_j(L_q(A(r,k)))}{n}-\frac{j-1}{r}\right)=\frac{k\log_2 M(r,k)}{r^{k-1}\log_2 r}>C.$$

Thus all sufficiently large primes q give the required strict inequalities. Infinitely many such primes also satisfy $kq\ge f(r)$, proving the assertion with arbitrarily large degrees. □

The high-dimensional base already illustrates why a constant independent of the uniformity is too restrictive. It is also cross-edge-free. To see this, fix a grid point v and a coordinate line e not containing it. A neighbor of v differs from v in exactly one coordinate. If e contained two such neighbors, both could differ from v only in the varying coordinate of e, which would force $v\in e$. Hence e contains at most one neighbor of v. Applying (18) directly to the base gives counterexamples to a uniform constant with fixed degree $k>C$ and large r. The lift in Theorem 1.3 strengthens the degree quantifier, but it generally loses cross-edge freeness, as explained in Remark 3.3.

There is a distinct limitation when the uniformity is fixed. Within the lifting mechanism of Theorem 2.2, the multiplicity of extremal supports cannot make the coefficient of $1/d$ unbounded. The following elementary estimate records this boundary of the construction.

Proposition 4.1. *Let A satisfy the assumptions of Theorem 2.2, and let M be its number of strong independent sets of size N/r. Then*

$$\frac{k\log_2 M}{N}\le\frac{k(r-1)}{k+r-1}<r-1.$$

Proof. Let B be the edge–vertex incidence matrix of A. A strong independent set of size N/r has incidence vector $x \in \{0,1\}^N$ satisfying $Bx = \mathbf{1}$. The diagonal entries of $B^{\mathsf{T}}B$ are k. Its off-diagonal entries are 0 or 1 by linearity, and each row has exactly $k(r-1)$ off-diagonal entries equal to 1. Hence

$$\operatorname{tr}(B^{\mathsf{T}}B) = Nk, \qquad \operatorname{tr}\big((B^{\mathsf{T}}B)^2\big) = Nk(k+r-1).$$

Cauchy's inequality applied to the nonzero eigenvalues gives

$$\operatorname{rank} B \geq \frac{Nk}{k+r-1}.$$

The solution space of $Bx = \mathbf{1}$ therefore has dimension at most $N(r-1)/(k+r-1)$. An affine subspace of dimension s contains at most 2^s vectors in $\{0,1\}^N$, because there is a set of s coordinate positions whose projection is injective on that subspace. Thus $\log_2 M \leq N(r-1)/(k+r-1)$, which proves the stated estimate. □

Theorems 1.1 and 1.3 leave open the optimal second-order bound at a fixed uniformity. Proposition 4.1 only limits the extremal-support lifting mechanism; it is not an upper bound for arbitrary regular linear hypergraphs. The asymptotic transfer theorem also requires the base to attain the extremal strong-independent-set size N/r. If this condition fails, configurations with all fibres nonempty over some base edge can contribute on a different exponential scale, and the present argument does not identify the leading term.

Declaration of AI use. Generative AI tools assisted with exploring constructions, checking calculations, and preparing the exposition.

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